

1 Heegaard Diagrams of 3-Manifolds

Example 1 ($L(p, q) := S^3 / [(\mathbf{w}, \mathbf{z}) \sim (\xi_p \mathbf{w}, \xi_q \mathbf{z})]$). Consider the standard Heegaard decomposition of S^3 given by

$$S^3 = \underbrace{\{|z| \geq |w|\}}_{h^1} \cup \underbrace{|z| = |w|}_{\mathbb{T}^2} \cup \underbrace{\{|w| \geq |z|\}}_{h^2} \quad (1)$$

Then the action of $\mathbb{Z}/(p)$ which I will denote by f preserves the handlebodies in this decomposition alongside the torus $|z| = |w|$. This gives us

$$L(p, q) = h^1 / f \cup_{\mathbb{T}^2 / f} h^2 / f \quad (2)$$

This is useless unless we can describe what these sets look like, notice that if we can describe $\mathbb{T}^2 / f$, then we have also described the handlebodies, since these can be expressed as $\mathbb{T}^2 / f \times I$. Computing for $\mathbb{T}^2$:

$$\mathbb{T}^2 / f = \frac{\mathbb{T}^2}{(x, y) \sim \left(x + \frac{1}{p}, y + \frac{q}{p}\right)} = \frac{\mathbb{R}^2 / (\mathbb{Z}^2)}{\left(\frac{1}{p}, \frac{q}{p}\right)\mathbb{Z}} = \frac{\mathbb{R}^2}{(0, 1)\mathbb{Z} \oplus \left(\frac{1}{p}, \frac{q}{p}\right)\mathbb{Z}} \cong \mathbb{T}^2 \quad (3)$$

Thus equation (2) also specifies a Heegaard Decomposition. Using the basis computed in equation (3) we take the corresponding generators for $H_1(\partial h^j)$. Now $(0, 1)\mathbb{Z}$ is the meridian for h^1 and thus also h^1 / f , same for $(1, 0)$ as the meridian of $h^2, h^2 / f$. The meridian of h^2 is thus glued to $p(1/p, q/p) - q(0, 1)$. This gives the heegaard decomposition on $\mathbb{T}^2$ as the curves given by $(\alpha, p\beta - q\alpha)$, where $\alpha, \beta \in H^1(\mathbb{T}^2)$ are the homology basis of the quotient computed in equation (3).

Example 2 ($\mathbb{T}^3$ with the flat metric). Model $\mathbb{T}^3$ as $[0, 2\pi]^3$, then consider

$$f(x) = \sum_1^3 \frac{1}{2} \cos(x^j) + \frac{3}{2} \quad (4)$$

I claim that this is a self-indexing morse function, checking this

$$-\nabla f = \frac{1}{2} \sin(x^j) dx_j \implies (-\nabla f)^{-1}(0) = \{(s, t, r) \mid s, t, r \in \{0, \pi\}\} \quad (5)$$

And at such points we have

$$\text{Hess}(f) = \frac{1}{2} \text{Diag}(-\cos(x^j)) \quad (6)$$

This verifies f is morse, to see its self indexing note that the index is the number of occurences of 0 in the critical point vector.

Now we solve for a gradient flow of f , from the above computations

$$\dot{x}^j = \frac{1}{2} \sin(x^j) \quad (7)$$

We can apply separation of variables to $y' / \sin y = \frac{1}{2}$ this gives us $\int \frac{1}{\sin(x^j)} dx^j = \frac{t}{2} + C$ integrating gives us

$$\log(\tan(x^j/2)) = \frac{t}{2} + C \quad (8)$$

and thus

$$x^j = 2 \arctan(Ce^{t/2}) \quad (9)$$

Using the initial condition $x^j(0), C = \tan(x^j(0)/2)$ so that

$$x^j = 2 \arctan(\tan(x^j(0)/2)e^{t/2}) \quad (10)$$

In order to compute the